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MILITARY SPECIFICATION

CORD, ELECTRICAL; AUDIO, SUBMINIATURE (RETRACTILE AND STRAIGHT)

1. SCOPE

1.1 Scope. This specification covers subminiature audio helical retractile cords and straight cords with either stranded wire or tinsel wire conductors, for accessories suitable for use with microphone, telephone, switchboard, and associated communication equipment. These cords are intended for operations at audio frequencies and temperatures from -55°C to +85°C with a high degree of flexibility. Each retractile cord is designed for extension in normal service up to 6 times its nominal retracted helical length.

1.2 Classification. The types of cords covered by this specification are classified as follows:

Type I - Retractile cords with stranded wire conductors.

Type II - Retractile cords with tinsel wire conductors.

Type III - Straight cords with stranded wire conductors.

Type IV - Straight cords with tinsel wire conductors.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

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SPECIFICATIONS

MILITARY

MIL-C-572	- Cords, Yarns, and Monofilaments, Organic Synthetic Fiber.
MIL-I-631	- Insulation, Electrical, Synthetic Resin Composition, Non-Rigid.
MIL-W-3795	- Wire, Electrical (Tinsel).
MIL-I-3930	- Insulating and Jacketing Compounds, Electrical (For Cable, Cord and Wire).
MIL-C-12000	- Cable, Cord and Wire, Electric; Packaging of.
MIL-C-45662	- Calibration System Requirements.
MIL-C-55442	- Cable Assemblies and Cord Assemblies, Packaging of.

STANDARDS

FEDERAL

Test Method STD-228	- Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104	- Limits for Electrical Insulation Color.
MIL-STD-105	- Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

RURAL ELECTRIFICATION ADMINISTRATION (REA) SPECIFICATIONS

PE-23	Fully Color-Coded, Polyethylene or Crystalline Propylene-Ethylene Copolymer-Insulated, Double Polyethylene Jacketed Telephone Cables for Direct Burial.
PE-210	Crystalline Propylene-Ethylene Copolymer Raw Material.

(Applications for copies should be addressed to the United States Department of Agriculture, Rural Electrification Administration, Washington, D. C. 20250.)

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AMERICAN SOCIETY OF TESTING AND MATERIALS

B33-63 Standard Specification for Tinned Soft or Annealed Copper Wire for Electrical Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 First article. The cords furnished under this specification shall be products which have been tested to and have passed the first article inspection requirements specified in 4.4. First article approval is valid only on the contract under which it is granted, unless extended by the government to another contract. (See 6.3.)

3.2 Conductors.

3.2.1 Stranded wire. Each stranded wire shall be the equivalent of No. 26 AWG and shall consist of 26 strands of No. 40 AWG wire having a cross-sectional area of 9.9 circular mils and stranded with a 0.25 inch nominal, 0.38 inch maximum, unidirectional left-hand lay. The diameter of the stranded conductor shall be approximately 0.019 inch. Each strand shall be tinned, soft or drawn-and-annealed cadmium-copper alloy having a 1 percent nominal cadmium content and a minimum conductivity of 85 percent.

3.2.1.1 Tin-coating. Each strand shall be tin-coated with commercially pure tin. The tin-coating, continuity, adherence and finish shall meet the requirements of ASTM Standard B33-63. (See 4.6.2.1.)

3.2.1.2 Stranded wire tensile strength and elongation. When tested as specified in 4.6.2.2, the tensile strength and elongation of the coated strands, removed from the wires of insulated conductors prior to cabling, shall be as follows:

Tensile strength	Elongation in 10 inches
40,000 psi (min.)	13 percent (min.)

3.2.1.3 Stranded wire resistance. Each strand shall have a direct current resistance of not more than 1.27 ohms per foot. The d.c. resistance of the No. 26 stranded conductor shall be not more than 0.0556 ohms per foot. Resistance values should be measured at or corrected to 20 degrees C when tested in accordance with 4.6.2.3. (See 6.2.)

3.2.2 Tinsel wire. Tinsel wire shall meet the requirements of MIL-W-3795, type IV. (See 6.2.)

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3.3 Insulation. The insulation for both stranded and tinsel wire conductors shall be a crystalline ethylene-propylene copolymer thermoplastic meeting the requirements of REA Specification PE 210. Properties of insulation removed from insulated conductors of finished cord shall meet the requirements of REA Specification PE 23, except that the cold bend shall be performed at -55°C. When tested in accordance with paragraph 4.6.4.2 (heat shock), there shall be no evidence of cracking. (See 4.6.3 and 4.6.4.)

3.3.1 Insulation wall thickness. Wall thickness at any section of insulation shall be nominal, 0.007 inch; minimum, 0.004 inch.

3.3.2 Dimensions of insulated conductors. The diameter of the insulated stranded and tinsel conductors shall be 0.035 inch, +0.004, -0.002 inch.

3.3.3 Insulation colors. Color of insulation shall conform to MIL-STD-104, class 1. The colors of individual insulated conductors in a core complement shall be in the following sequence:

1. Black	6. Yellow
2. White	7. Blue
3. Red	8. Purple
4. Green	9. Brown
5. Orange	10. Gray

3.4 Fibers and yarns. All synthetic yarns and fibers employed as servings, binders, fillers, cores for tinsel strands and wires, and staycords shall be as specified herein.

3.4.1 Fillers. Filler yarn shall be used between insulated conductors to form an essentially circular cross-section core and provide a slippage between the insulated wires. Fillers shall be a yarn conforming to MIL-C-572 type PSTR (polyester) or type CTA (cellulose triacetate), or a polypropylene yarn or twisted polypropylene film.

3.4.2 Staycord. Staycord shall consist of a polyaryl-amide fiber conforming to MIL-C-572, type PAA. The cord shall have a minimum breaking strength of 20 pounds for cords of 2, 3 and 4 conductors, and thirty pounds for cords of 5 to 10 conductors. The filaments of the staycord shall be bonded together with a fungus-inert adhesive in order to facilitate knotting during termination of the finished cord.

3.4.2.1 Center core. If necessary to provide a center core larger in diameter than the staycord, a filler material as specified in 3.4.1 shall be served around the staycord.

3.5 Shield. Shielding shall be served tinsel strands. The ribbon comprising the strands shall be silver or tin-coated cadmium-copper of 0.0095 inch nominal width and 0.0009 inch nominal thickness. A strand shall consist of a single

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ribbon, spiral-wrapped 54 turns per inch, nominal, around a cotton thread, or a polyaryl-amide thread conforming to MIL-C-572, type PAA. The shield shall consist of 10 tinsel strands laid adjacent to each other, and spiral-wrapped around the insulated conductor in an even helical right-hand lay, with a minimum of 60 turns per foot. At no cross-section should there be less than 8 strands in a turn. Shield coverage shall be 80 percent minimum, as calculated in accordance with paragraph 4.6.5. An overlapping spiral-wrap of 0.0005 inch nominal thickness polyester tape per MIL-I-631, type G, shall be applied in a right-hand lay over the shield. Another tape may be placed between the insulation and the shield. The diameter of the taped, shielded insulated wire shall be approximately 0.052 inch.

3.5.1 Shield resistance. The direct current resistance of a shield strand prior to application on the insulated conductor shall be 2.50 ohms per foot maximum, measured at, or corrected to, 20°C. (See 6.2.)

3.6 Jacket. The jacket shall be a smooth, dense, uniform black elastomer free of surface irregularities. Jacket dimensions shall be as specified in table II. The wall thickness shall be measured on the straight ends. The jacket material shall be a black vulcanized rubber compound based on one of the following:

(a) Styrene butadiene rubber (SBR) or blend of SBR and polyisoprene rubber, not exceeding 35 percent natural and/or synthetic polyisoprene by weight.

(b) A blend of SBR and ethylene-propylene terpolymer (EPDM), with 25 percent EPDM by weight, minimum.

(c) Ethylene-propylene copolymer (EPM) rubber or ethylene-propylene terpolymer (EPDM) rubber.

3.6.1 Jacket material properties. (See 4.6.6.) Vulcanized rubber jacket materials shall meet the requirements of MIL-I-3930, type JS-L, except as specified in table I. The actual values shall be recorded and forwarded with first article sample test data. (See 3.1.) Unless otherwise indicated, specimens shall be taken from the straight section of finished cord; for first article qualification, cords may be made up with extra-long straight sections.

3.6.2 Anti-deterioration agents. The rubber compound shall contain anti-oxidants, anti-ozonants, and sunproofing waxes of the types and amounts necessary to provide long term protection, yet not interfere with adhesion to subsequent rubber strain-reliefs, seals, or crotch-splices.

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Table I. Properties of jacket material.

Property	Value
Unaged	
Ultimate elongation, minimum average percent	175
with no value less than, percent	150
Tear resistance, minimum lb./in.	15
Cold tension recovery at -55°C, minimum percent	20
Hot tension recovery at 85°C, minimum percent	75
Free sulfur, maximum percent	0.25
Aged	
Oxygen bomb, 95 ±1 hour at 70°C -	
Change in tensile strength, maximum percent	25
Change in elongation, maximum percent	35
Air oven, 168 ±1 hour at 100°C -	
Change in tensile strength, maximum percent	10
Change in elongation, maximum percent	50
Ozone resistance after oven aging	No cracks

3.7 Cord construction.

3.7.1 Cord construction parameters. The core shall consist of a staycord in the center around which the insulated conductors and fillers are twisted in a right-hand lay. Powdered soapstone (lubricant) shall be applied to the completed core just prior to jacketing the cord. For a retractile cord, the helical portion shall have a left-hand lay with adjacent turns contiguous and a straight length of cord at each end tangent to the coil. The cord construction parameters consisting of the number of conductors, number of shielded conductors, core diameter, core lay, jacket wall thickness, jacket diameter, and, for retractile cords, the helix outside diameter shall be in accordance with table II.

3.7.2 Cord procurement data (see 6.1). The contract or order shall specify the type of cord (see 1.2), number of insulated conductors (shielded and unshielded), and total length of cord. For retractile cords, the length of helix and/or number of turns, length of straight ends, and overall length of cord shall be specified. In the event of any conflict between requirements of this specification and the contract or order, the latter shall govern.

3.8 Requirements of the completed cord. The completed cord shall meet the requirements specified in table III when tested under the applicable test paragraph, except that the extension and/or retraction requirements and tests of 4.6.7, 4.6.7.2, 4.6.7.3, 4.6.7.4, 4.6.9, 4.6.11 and 4.6.12 do not apply to straight cords.

3.9 Workmanship. Cords shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques. The cords shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

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Table II. Construction dimensions (*) for straight and rectractile cords.

Total no. of conductors	No. of conductors		Core dia. nom.	Core lay		Jacket wall thickness		Jacket dia.		Helix O.D. (for retractile cords)	
	Unshielded	Shielded		nom.	max.	nom.	min.	nom.	tol. (±)	nom.	max.
2	2	0	0.083	7/16	9/16	0.026	0.016	0.135	0.006	16/32	20/32
3	3	0	.085	5/8	3/4	.040	.024	.165	.007	18/32	23/32
3	2	1	.095	7/16	9/16	.035	.021	.165	.007	18/32	23/32
4	4	0	.090	5/8	3/4	.037	.022	.165	.007	18/32	23/32
4	3	1	.095	5/8	3/4	.035	.021	.165	.007	18/32	23/32
5	5	0	.102	5/8	3/4	.031	.019	.165	.007	18/32	23/32
5	4	1	.098	5/8	3/4	.033	.020	.165	.007	18/32	23/32
6	5	1	.110	5/8	3/4	.027	.016	.165	.007	18/32	23/32
7**	7	0	.116	3/4	7/8	.037	.022	.190	.008	21/32	25/32
8**	7	1	.125	3/4	7/8	.032	.019	.190	.008	21/32	25/32
9**	7	2	.130	3/4	7/8	.030	.018	.190	.008	21/32	25/32

* All dimensions are in inches.

**Fillers in valleys are not required.

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Table III. Cord requirements.

Test paragraph	Property	Value
4.6.7	Cord retractility	Contiguous
4.6.7.1	Breaking load, minimum pounds	70
4.6.7.2	Cord extension (7:1) force, maximum ounces	32
4.6.7.3	Retraction after static vertical loading, percent of original length:	
	Stranded wire cord, maximum percent	120
	Tinsel wire cord, maximum percent	110
4.6.11	Stretch cycling (6:1) extension):	
	(a) Variation in resistance during cycling, maximum percent	5
	(b) Increase in resistance during or after cycling, maximum percent	10
	(c) Retraction after cycling, maximum percent of original length	120
4.6.10	Bending flex (180°):	
	Stranded wire cord, minimum number of cycles	5,000
	Tinsel wire cord, minimum number of cycles	15,000
4.6.7.4	Low temperature (-55°C) extension and retraction:	
	(a) Cord extension (4:1) force, maximum ounces	35
	(b) Retraction, percent of original length:	
	Stranded wire cord, maximum percent	160
	Tinsel wire cord, maximum percent	125
4.6.8	Cold bend (-55°C)	No cracks
4.6.9	Low temperature (-55°C) rapid extension	No cracks
4.6.12	Retraction at 85°C, maximum percent of original length	275
4.6.13	Voltage withstand, as received, test volts	1,000, without breakdown
4.6.14	Insulation resistance, as received, minimum megohms	1,000
4.6.15	Continuity	No opens
4.6.16	Humidity cycling:	
4.6.13	(a) Voltage withstand, test volts	1,000, without breakdown
4.6.14	(b) Insulation resistance per finished cord length, minimum megohms	1,000
4.6.15	(c) Continuity	No opens
4.6.17	Simulated storage, 14 days/85°C:	
4.6.13	(a) Voltage withstand, test volts	1,000 without breakdown
4.6.14	(b) Insulation resistance, per finished cord length, minimum megohms	1,000
4.6.9	(c) Low temperature (-55°C) rapid extension	No cracks

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment and facilities. Inspection equipment and facilities shall be established and maintained in accordance with MIL-C-45662.

4.2 Classification of inspections. The examinations and tests of cords are classified as follows:

a. Materials inspection. (See 4.3.)

b. First article inspection. (See 4.4.)

c. Quality conformance inspection. (See 4.5.)

1. Inspection of product for delivery. (See 4.5.1.)
2. Inspection of preparation for delivery. (See 4.5.5.)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table IV, used in fabricating subminiature cords, are in accordance with the applicable referenced specification or requirements prior to such fabrication.

Table IV. Materials inspection.

Material	Requirement paragraph	Applicable specification
Tinsel wire	3.2.2	MIL-W-3795, type IV
Insulation	3.3	REA PE-210 and PE-23
Fillers, Center core	3.4.1, 3.4.3	- - - - - - -
Cellulose triacetate		MIL-C-572, type CTA
Polyester		MIL-C-572, type PSTR
Polypropylene		- - - - - - -
Staycords	3.4.2	MIL-C-572, type PAA
Polyester tape	3.5	MIL-I-631, type G
Jacket material	3.6.1 and table I	MIL-I-3930, type JS-L

4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the government. (See 6.3.) First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. (See 3.1.)

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4.4.1 Sample size and test routine. Twenty-five sample units shall be subjected to the tests specified in table V. These sample units shall be of sufficient length to perform the tests specified.

4.4.2 Submission of data for approval. Test results (and samples, when specified) shall be forwarded to the contracting officer for approval.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspection. Delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all cords of the same type (see 3.7) produced under essentially the same conditions and submitted for inspection at the one time.

4.5.1.2 Unit of product. The unit of product, for purposes of sampling, shall be each continuous length of cord as specified in the contract or order.

4.5.1.3 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan for the presented lot.

4.5.1.4 Selection of test specimens. Test specimens for inspection shall be taken from each unit of product which forms part of the sample.

4.5.1.5 Nonconforming inspection lots. Disposition of inspection lots found unacceptable under initial inspection shall be in accordance with MIL-STD-105.

4.5.1.6 Finished cords. A finished (or completed) cord is one of a definite length, or a continuous length of straight cord on a reel, spool or coiled, as specified in the contract or order, with all manufacturing operations completed.

4.5.2 Group A inspection. Group A inspection shall include the examinations and tests of table VI, subgroups I and II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.2.1 Subgroup I. For subgroup I, the acceptable quality level (AQL) shall be as specified in table VI and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup I tests may be performed in any order.

4.5.2.2 Subgroup II. For subgroup II, the AQL shall be as specified in table VI and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup II tests shall be performed in the order shown in table VI.

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Table V. First article inspection tests.

Paragraph	Property
3.2.1	Stranded wire construction, coating; tensile strength and elongation of stranded conductor.
3.2.1.3	Stranded wire conductor resistance.
3.2.2	Tinsel wire construction, properties.
3.3	Insulation properties.
3.3.1	Insulation wall thickness.
3.3.2	Dimensions of insulated conductors.
3.3.3	Insulation colors.
3.4	Fibers and yarns.
3.4.1	Fillers.
3.4.2	Staycord.
3.5	Shield wire; construction, coverage, outside diameter.
3.5.1	Shield resistance.
3.6	Jacket wall thickness and outside diameter.
3.6.1	Jacket material properties.
3.7	Cord construction.
3.8	Cord properties: Retractility, breaking load, cord extension (7:1) force, retraction after static vertical loading, low temperature extension and retraction, cold bend, retraction at 85°C, voltage withstand, insulation resistance, continuity, stretch cycling, bending flex, humidity cycling and simulated storage.

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Table VI. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (Percent defective) Major	AQL (Percent defective) Minor
Subgroup I				
Visual and mechanical				
Material and construction	3.2 thru 3.7 incl	4.6.1	1.0% for the subgroup	4.0% for the subgroup
Workmanship	3.9	4.6.1		
Subgroup II				
Electrical				
Voltage withstand	Table III	4.6.13	1% for the subgroup	*
Insulation resistance		4.6.14		
Continuity		4.6.15		

*All electrical defects are considered major.

4.5.2.3 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.3 Group B inspection. This inspection, including sampling, shall conform to table VII and to the procedures for small-sample inspection of MIL-STD-105, using special inspection levels. Group B inspection shall be performed on inspection lots that have passed group A inspection and on specimens selected from units of product that have been subjected to and met the group A inspection. The AQL shall be 6.5 (percent defective), applied individually to each subgroup of group B, and the special inspection level shall be S-2.

4.5.3.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3.2 Disposition of samples. Samples subjected to group B tests shall not be delivered on contract or order.

4.5.4 Group C inspection. This inspection shall consist of the tests specified in table VIII and shall be performed periodically as indicated in 4.5.4.1. Samples shall be selected from lots that have passed group A and B inspection.

4.5.4.1 Sampling for inspection. One sample shall be selected per each 5,000 feet of cord or fraction thereof, not to exceed a total of five samples per order. Samples shall be selected periodically through the life of the contract to assure representative results.

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Table VII. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Subgroup I		
Stranded wire		
Tin-coating	3.2.1.1	4.6.2.1
Tensile strength and elongation	3.2.1.2	4.6.2.2
Resistance	3.2.1.3	4.6.2.3
Subgroup II		
Cord		
Retractility	Table III, except as modified by 3.8	4.6.7
Breaking load		4.6.7.1
Extension force		4.6.7.2
Retraction after static vertical loading		4.6.7.3
Bending flex		4.6.10
Stretch cycling		4.6.11
Low temperature (-55°C) extension and retraction		4.6.7.4
Cold bend		4.6.8
Retraction at 85°C		4.6.12

Table VIII. Group C inspection.

Examination or test	Requirement paragraph	Test method paragraph
Low temperature rapid extension	Table III, except as modified by 3.8	4.6.9
Humidity cycling		4.6.16
Simulated storage		4.6.17

4.5.4.2 Noncompliance. No failures shall be allowed in group C inspection. If a sample unit fails to pass group C inspection, the supplier shall take corrective action on the material or process or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection that the original sample failed, at the option of the government). Groups A and B inspection may be reinstituted, however, final acceptance shall

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be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.5 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

4.6 Test methods.

4.6.1 Visual and mechanical inspection. The finished cords shall be given a visual and mechanical inspection for conformance to the applicable requirements of table VI. Visual and mechanical defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105.

4.6.2 Stranded wire.

4.6.2.1 Tin-coating. The tin-coating on the conductor strands and the continuity, adherence and finish shall pass the applicable tests specified in ASTM Standard B33-63. (See 3.2.1.1.)

4.6.2.2 Tensile strength and elongation. This test shall be performed in accordance with FED-STD-228, Method 3211. (See 3.2.1.2.)

4.6.2.3 Stranded wire resistance. The direct current resistance of conductors shall be measured after being insulated, but prior to cabling, in accordance with FED-STD-228, Method 6021. (See 3.2.1.3.)

4.6.3 Insulation and jacket tension recovery. (See 3.3 and 3.6.) The apparatus and procedure for tension recovery of insulation and jacket shall be in accordance with "cold-tension recovery" of MIL-I-3930, except as specified below.

4.6.3.1 Specimens. Specimens for tension recovery shall be tubular, taken from straight sections of the finished cord.

4.6.3.2 Procedure. The specimen shall be stretched at room temperature to 50 percent elongation and conditioned for 22-24 hours at the specified test temperature. At the end of the conditioning period, the specimen shall be released and the percent recovery shall be measured at the end of 1 minute.

4.6.4 Testing of insulation. (See 3.3.) Properties of the insulation as removed from finished cord shall be determined in accordance with applicable paragraphs of REA Specification PE-23, except as specified below.

4.6.4.1 Tensile strength and elongation. The speed of jaw separation shall be 2 inches per minute. Quality assurance testing at 20 inches per minute is permissible if no failures occur.

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4.6.4.2 Tubing requirements of insulation. To determine whether the extrusion process has caused strains in the insulation, a specimen of each insulated conductor, taken from the straight section of a finished cord, shall be looped back and then wound tightly on itself for 5 close turns, the ends securely taped, and the specimen placed in a circulating air oven maintained at 121° ±1°C, for a period of 60 -2 minutes. After removal from the oven, the specimen shall show no evidence of cracking of the insulation when examined under 3X magnification.

4.6.5 Shield coverage. (See 3.5.) Shield coverage shall be calculated as follows:

Percent coverage, K = (NPd/sin a) x 100

where:

N = number of tinsel strands per carrier
P = picks per inch
d = diameter of tinsel strand, in inches
a = angle of serving with axis of conductor

$$\text{Tan } a = 2\pi (D+d) \times \frac{P}{C}$$

where:

D = diameter under served shield, in inches
C = number of carriers

4.6.6 Testing of jacket material. (See 3.6.1.) Properties of the jacket material shall be determined in accordance with applicable paragraphs of MIL-I-3930 except as indicated herein.

4.6.6.1 Ozone resistance test. The specimen shall be a completed cord or helical section thereof. The specimen shall be air-oven aged at 70° ±1°C for a period of 168 ±1 hour in an unstressed condition and allowed to cool to room temperature prior to placement in the ozone chamber. A inch (minimum) section of the helical section shall be stretched 5:1 and maintained in this elongated condition while in the ozone chamber and during examination for cracks at room temperature.

4.6.7 Cord retractility. Prior to meeting any extensile and retractile requirements, the completed cord shall be conditioned by extending the coiled portion to 5 times its retracted length and allowing it to retract freely for a total of 6 cycles. After this conditioning, the cord shall be placed on a horizontal surface and the coils examined for contiguity. (See table III.)

4.6.7.1 Breaking load. The test specimen shall be a straight cord or a retractile cord, as applicable. The straight cord shall be approximately 1 foot long; the retractile cord (or section thereof) shall contain at least 5 helical turns and shall be approximately 1 foot long in extended length including a straight section. For retractile cords with less than 5 turns, the entire cord shall be used. The breaking load shall be determined on a tensile

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testing machine, power driven, and preferably of the pendulum type. The test shall be made at the rate of 12 inches per minute. The conductors of the cord may be connected in series to an indicating or signaling device or automatic stop, or both. The arrangement shall be such that an indication will be given when there is a disruption of electrical continuity due to the breaking of the conductors of the cord under test. The minimum load shall be as specified in table III.

4.6.7.2 Cord extension force. The finished cord, or helical portion thereof, in a horizontal position shall be fastened at one end, with all coils contiguous. The length of the helical portion shall be measured, and a tensile force applied to the free end of the cord, sufficient to extend the helical portion 7 times its retracted length. The maximum force required for this extension shall be as specified in table III.

4.6.7.3 Retraction after static vertical loading. At room temperature, the finished cord shall be secured at one end and hung vertically with a weight attached to the lower end, sufficient to extend the helical portion of cord to approximately 250 percent of its original horizontal retracted length. After 48 hours, the weight shall be removed and the cord placed on a horizontal surface. This surface shall be tapped to reduce frictional restraint. Thirty minutes after being placed on the horizontal surface, the helical portion of the cord shall be measured to determine percent retraction of original length. (See table III.) Percent retraction shall be calculated as follows:

$$\text{Percent retraction} = \frac{\text{Retracted length after extension}}{\text{Original retracted length}} \times 100$$

4.6.7.4 Low temperature (-55°C) extension and retraction. A 6 inch helical portion of a completed cord with all coils contiguous shall be placed in a horizontal position within an environmental test chamber maintained at -55° +1°C, fastened at one end, and conditioned for not less than 20 hours prior to testing. If the helical portion of the finished cord is less than 6 inches long, the complete cord shall be used and measurements made of the helical portion. At the conditioning temperature, a force shall be applied to the free end of the cord sufficient to extend the helical portion 4 times its retracted length within 5 seconds. The maximum force shall be as specified in table III. Upon release of the force and while still at -55°C, within 30 seconds after release, the helical portion of the specimen shall retract to not more than the percent of its original length as specified in table III. Frictional restraint shall be minimized by tapping the surface on which the cord rests. Upon visual examination at room temperature, the jacket, insulation, and conductors in both extended and retracted states shall be free of cracks and other defects.

4.6.8 Cold bend. Specimens shall be taken from the straight portions of the cords. The specimens from any one cord shall be divided, one part for checking the finished cord as a whole (jacketed cord) and the other for checking

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insulation, for compliance with table III. The part selected for checking insulation shall have the insulated conductors removed for test; shielding, if any, shall be removed. Each insulated conductor shall be tested. The test specimens shall be attached to mandrels of diameters equal to the diameters of the specimens, and shall be suspended vertically with lower ends weighted sufficiently to keep the specimens taut and to permit bending them without handling. The mandrels and specimens shall be placed for at least 24 hours in a cold chamber maintained at -55° ±1°C. While at this temperature, the specimens shall be bent for 5 close turns around the mandrels at the rate of approximately 1 turn per second. After removal from the cold chamber, the specimens shall be examined under 3X magnification for cracks, breaks, or other faults. Jacketed specimens shall be stripped and the insulation of the conductors examined.

4.6.9 Low temperature rapid extension. The test specimen shall be a completed cord if the helix is not greater than 8 inches long, or portion thereof including one straight end and at least 8 inches of the helical section. The specimen, with all coils contiguous, shall be placed in a horizontal position within a cold chamber and conditioned for 7 days at -55° ±1°C. While at -55°C, the specimen shall be secured at one end so that the cord hangs vertically, with a straight section at the bottom. A 1 pound (minimum) weight shall be attached to the free straight end and then allowed to fall. The distance of fall shall be sufficient to extend the helical portion of the cord to 4 times its retracted length. After removal from the test chamber, the cord shall be examined under 3X magnification for cracks or other defects in the jacket and conductor insulation. (See table III.)

4.6.10 Bending flex. The test specimen shall be a finished straight cord (or convenient length thereof), or a finished retractile cord, as applicable. The cord flexing machine shall contain a rotating actuating arm whose normal zero position is vertical. The arm shall rotate about a horizontal axis, the outer end traversing a circular arc in a vertical plane. The arm shall rotate 90 degrees from the zero position to a position 90 degrees from the zero position in the opposite direction, and then reverse its direction and rotate back to zero position, thus completing a cycle. The arm shall be rotated at approximately 14 cycles per minute. Two adjustable-position 0.300-inch diameter mandrels shall be mounted in the machine with their axis perpendicular to the plane of rotation of the actuating arm. The plane through the axis of the two mandrels shall be horizontal and shall pass through the axis of rotation of the actuating arm. The mandrels shall be placed so that the axis of rotation lies midway between the mandrels. With the fixture positioned vertically, the specimen shall be clamped 2 inches above the helix, so that the straight end protrudes above the clamping device and passes between the mandrels with some slight clearance. The conductors and shields shall be wired in series and connected to the electrical test apparatus as indicated by the schematic in figure 3. (A cord identical to the one being tested is also wired

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in series, and connected as the reference leg of the Wheatstone Bridge.) A 5 pound weight shall be attached to the 10th convolution of the helix (counting from the clamped end down). If the helix has less than 10 turns, the weight shall be clamped to the lower straight end, as near as possible to the helix. Figures 1 and 2 show the clamping and mandrel arrangements, with the 10 convolutions straightened by the weight. The test weight shall be restricted from twisting (to simulate service conditions) by a slack anchor attached to the weight. The electrical test circuit is energized and meter null adjustments made with the variable resistor (figure 3). A mechanical counter shall be used to record the number of cycles. The meter relay, figure 3, shall be set to trip at a fluctuation of 50 milliohms (±25 milliohms from the null point) in the resistance of the specimen, terminating the test. Tests on tinsel wire conductors shall be discontinued, if no failure occurs, at the number of cycles specified in table III. Tests on stranded wire conductors shall be run until failure and the number of cycles recorded, for comparison with the requirement in table III.

4.6.11 Stretch cycling. The test specimen shall be the finished cord, or, for very long cords, at least an 8 inch helical section. The resistance (R_i) of each conductor shall be measured prior to cycling. The helix shall then be extended to 6 times its retracted length and allowed to retract. This stretch cycle shall be repeated 60,000 times at a rate of not more than 40 cycles per minute. At the completion of 60,000 cycles, several additional cycles shall be run; during this period, the conductor resistance shall be monitored continuously.

(a) The percent variation in resistance during these cycles shall be calculated as follows:

$$\text{Percent variation} = \frac{R_v}{R} \times 100$$

Where R = The conductor resistance in ohms at the start of the final cycling.

R_v = Difference between highest and lowest resistance values of the conductor during the cycling.

(b) The percent increase in conductor resistance due to cycling shall be calculated as follows:

$$\text{Percent increase} = \frac{R_m - R_i}{R_i} \times 100$$

Where R_i = The conductor resistance in ohms before cycling.

R_m = Maximum resistance of the conductor during cycling or at completion of cycling, whichever is higher.

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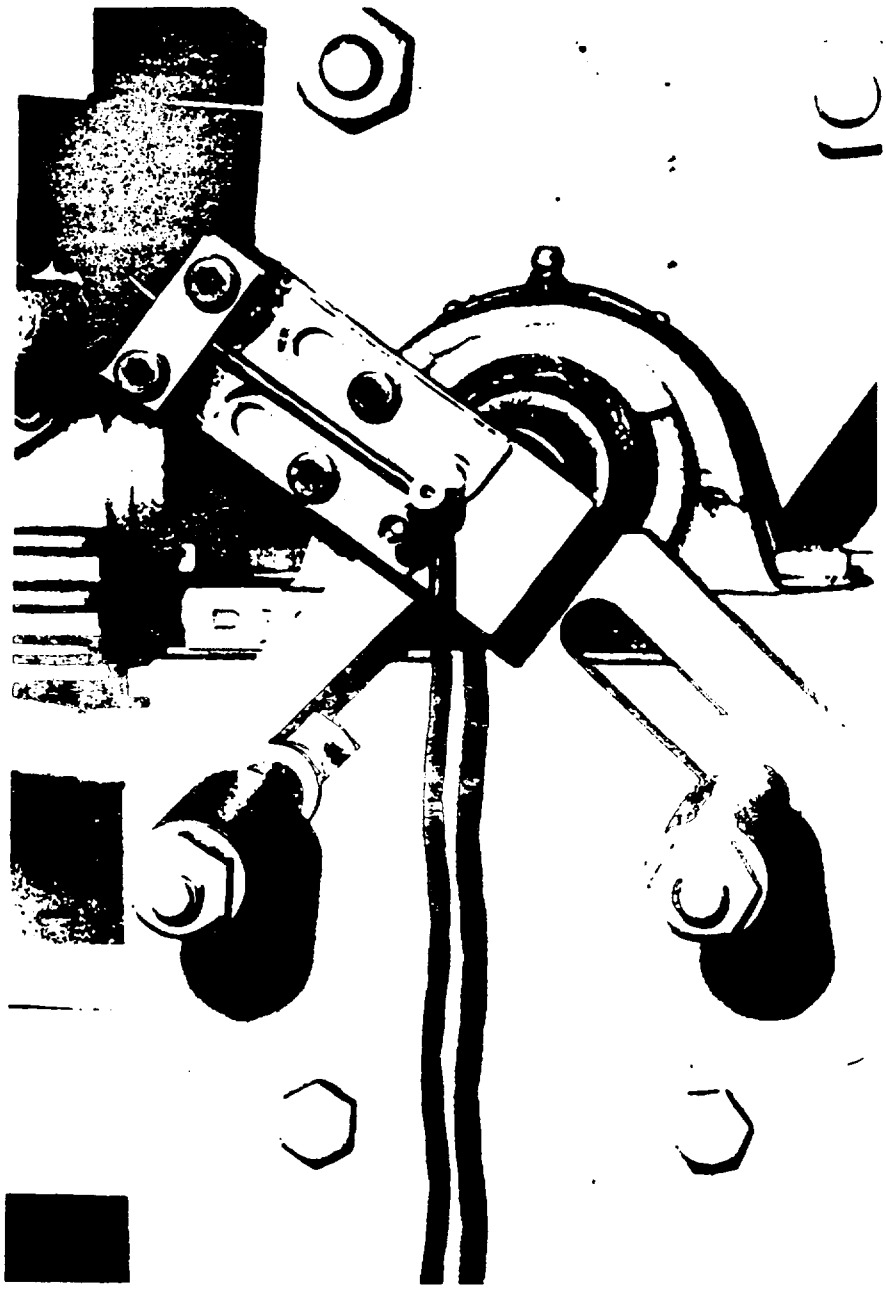


FIG. 1 - Bend flex tester (close-up).

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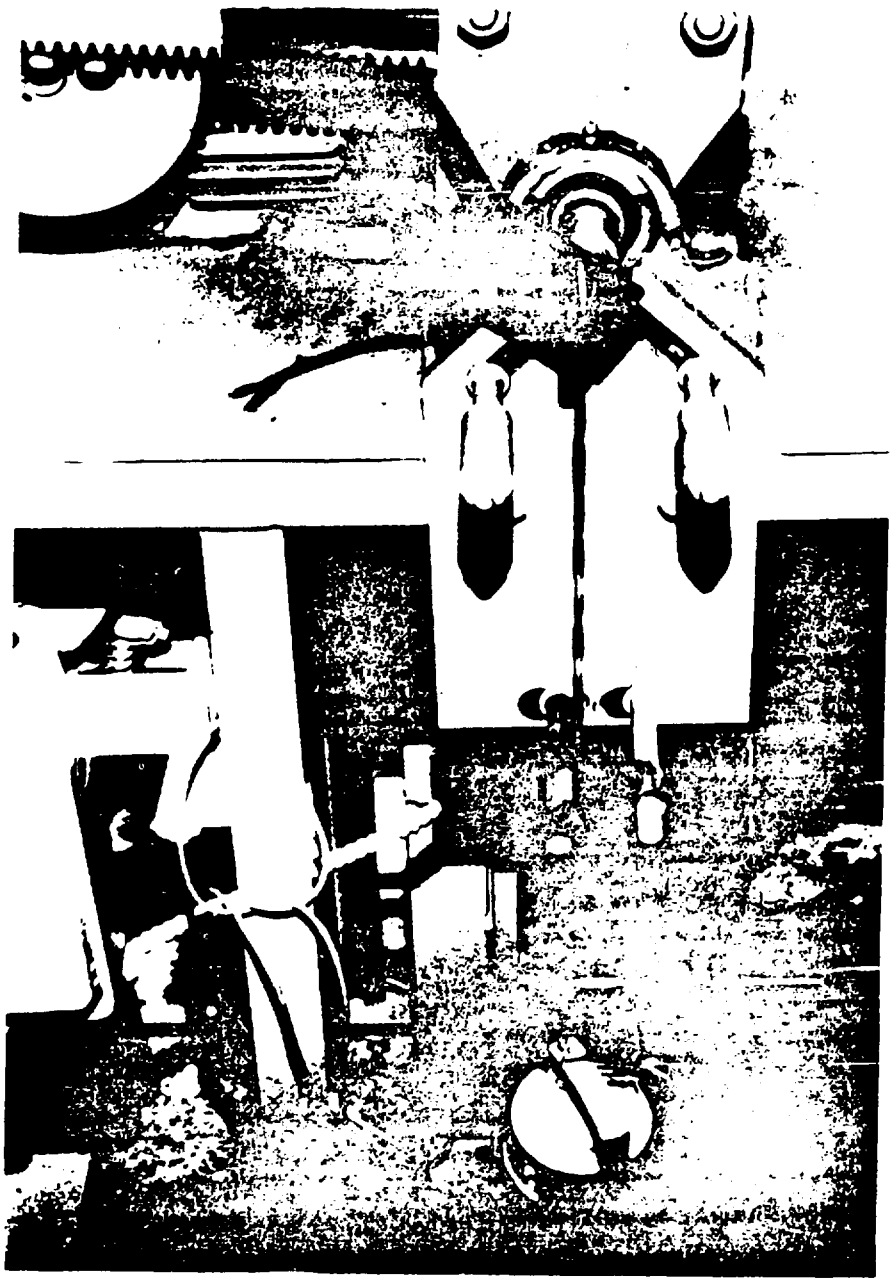


FIG. 1 - Bend flex tester.

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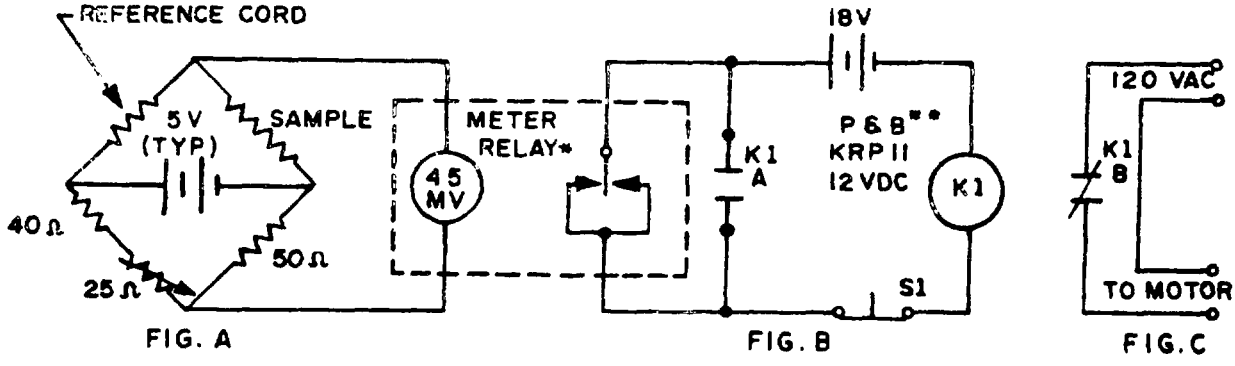


FIG. A FIG. B FIG. C

*Calder Bros. & Thompson Ltd., London, Ont. Model SP6W5, Zero Center, Approximate Full Scale Sensitivity 45 mv, or equal.

**Potter & Brumfield, or equal.

Figure A shows the sample cord connected as one arm in a Wheatstone Bridge. The opposite leg (reference cord) is an identical cord used to balance out temperature fluctuations in the sample. The meter relay is set to trip at a fluctuation of ±25 milliohms.

Upon tripping, the control relay K1 is pulled in, normally open contacts K1A are closed, and K1 latches in the energized position until release switch S1 is depressed as shown in Figure B.

Figure C shows the second pole of relay K1 as it is used to control the flex tester motor.

The power supply is shown as a battery for simplicity.

Fig. 3 - Basic circuit used for 180 degree bending flex control.

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The maximum percent variation in the resistance of any conductor shall be as specified in table III. At the completion of the cycling test, the maximum percent increase in resistance of any conductor over the resistance before cycling shall be as specified in table III. All resistance measurements shall be at or corrected to 20°C. The helical portion of the test specimen shall be measured for percent retraction within 1 hour after completion of the cycling. Measurements shall be made on a horizontal surface, as specified in paragraph 4.6.7.3. The maximum percent retraction shall be as specified in table III.

4.6.12 Retraction at 85°C. The test specimen shall be a completed cord. With the helical section extended to 3 times its retracted length, the cord shall be conditioned in a circulating air oven maintained at 85° ±1°C for 20 hours. At the end of this period and while in the oven at 85°C, the helical section shall be extended further, to 5 times its retracted length, and returned to its original retracted length. The extension and retraction shall be repeated for a total of 3 cycles. The cord shall then be suspended vertically and measured for retraction at the end of 15 seconds, the percent retraction (calculated on the basis of original horizontal retracted length) shall meet the requirements specified in table III.

4.6.13 Voltage withstand. The finished retractile cord, or convenient length of straight cord, as applicable, shall be immersed in tap water, at room temperature, for at least 9 hours prior to testing. During the test, neither the extreme ends of the cord nor the bared conductor wires shall be immersed in water.

4.6.13.1 High-voltage source. The voltage withstand test shall be made with an alternating potential of commercial power frequency, having a wave form as nearly sinusoidal as practicable. The testing transformer and power source shall be capable of supplying no less than 3 kilovolt-amperes.

4.6.13.2 Voltage measurements. The testing voltage may be measured by any approved method which gives a root-mean-square value, preferably by means of a voltmeter connected to a special voltmeter coil in the high-voltage winding of the testing transformer or to a separate step-down transformer. A voltmeter on the low-voltage side of the transformer is satisfactory, provided the ratio of transformation does not change under any test condition.

4.6.13.3 Procedure. An alternating potential as specified in table III, shall be applied between two terminals, one being each conductor in turn, and the other being all the remaining conductors tied together in electrical contact with the water. The initially applied voltage shall be not greater than 600 volts, and the rate of increase shall be approximately uniform and not over 100 percent in 10 seconds, nor less than 100 percent in 60 seconds. The specified test voltage shall be applied for not less than one minute unless breakdown occurs prior to that.

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4.6.14 Insulation resistance. The insulation-resistance measurements shall be made on sample units which have successfully passed the voltage withstand test. (See 4.6.13.) The apparent insulation resistance shall be measured as soon as practicable after the voltage withstand test. The leakage current shall be measured after a 1-minute electrification with a direct-current potential of not less than 100 volts nor more than 500 volts. The polarity of the conductor wire shall be maintained negative with respect to the water. The insulation resistance of each cord shall be measured as follows: A direct current potential shall be applied between two terminals; (a) one terminal shall be all the conductors and shield wires tied together, and the other shall be the jacket surface which is in electrical contact with the water; (b) one terminal shall be each conductor wire in turn, and the other shall be all the remaining conductors plus all shield wires tied together in electrical contact with the water. If measurements are made at lower than 20°C, the measured value of each resistance shall be corrected to 20°C. The minimum resistance value per cord shall be as specified in table III.

4.6.15 Continuity. Each conductor and shield of each length of finished cord or cord on reel, as applicable, shall be tested for continuity. The maximum test voltage shall be 10 volts. (See table III.)

4.6.16 Humidity cycling. The test specimen shall be a finished cord which has passed the voltage withstand and insulation resistance tests as specified in table III. The cord shall be placed in a humidity chamber, and the ends of the cord brought out through the walls of the chamber for electrical measurements. The humidity cycle shall be in accordance with MIL-STD-202, Method 106, except Step 7. Voltage withstand and insulation resistance shall be measured before start of the environmental cycle and during the 10th cycle, after Step 6, at room temperature and 90-100 percent relative humidity, with the specimen in the chamber, in accordance with paragraphs 4.6.13 and 4.6.14, and shall meet the requirements of table III. The specimen shall then be removed from the chamber and subjected to the bending flex test of paragraph 4.6.10 to determine compliance with table III.

4.6.16.1 Alternate humidity cycle. An alternate method of humidity cycling is as follows:

Step A: 16 ±1/2 hours at 65°C/90-100 percent RH

Step B: 8 ±1/2 hours at room temperature/90-100 percent RH

Voltage withstand and insulation resistance measurements shall be made before start of the enviromental cycle and after 4 but not more than 6 hours have elapsed in Step A of the 10th cycle, with the specimen still in the chamber.

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4.6.17 Simulated storage. The test specimen shall be a completed cord. The cord shall be aged in a circulating air oven for 336 ±2 hours at 85° ±1°C, then cooled at room temperature for a minimum of 24 hours. Effects of plasticizer migration and/or loss shall be determined by testing voltage withstand, insulation resistance, and low temperature rapid extension as specified in paragraphs 4.6.13, 4.6.14, and 4.6.9, and table III.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with MIL-C-55442 for cords of specified length, and in accordance with MIL-C-12000 for bulk cordage on reels, spools or coiled. (See 6.1(c) and 6.1(d).)

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required. (See 3.7.)
(c) Length and quantity of finished cords to be delivered.
(d) Level of packaging and level of packing required. (See 5.1.)

6.1.1 Indirect shipments. The preservation, packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipments to the government, and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Typical resistance values. Typical approximate resistance values for conductors and shields in a finished straight cord 120 inches long, or a finished retractile cord with a 12-1/2 inch length of helix (rated for extension to 75 inches) and 6 inch long straight ends, are:

Stranded wire cord:	Conductor	- 0.44 ohms
	Shield	- 2.7 ohms
Tinsel wire cord:	Conductor	- 2.3 ohms
	Shield	- 2.7 ohms

6.3 First article inspection. Information pertaining to first article inspection of cords covered by this specification should be obtained from the procuring activity for the specific contracts involved. (See 3.1.)

6.3.1 First article inspection on repeat orders. First article inspection will not be required from a supplier who has manufactured the same type cord (same type of wire, number of conductors, number of shielded conductors, same materials, etc.) on a previous order or contract within 2 years of the current order or contract, and which passed first article inspection at that time. A supplier who has passed first article inspection on a cord of a type

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classification as specified in 1.2 will be considered qualified on cords of the same type classification providing these cords contain the same total number of conductors and shielded conductors as, or a lesser total number of conductors or shielded conductors than, the cords previously qualified and satisfactorily manufactured.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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